



Tire-making causing salmon deaths

Researchers have discovered a chemical in tires that kills coho salmon before the fish can spawn. <http://digit.in/may21-23>



New Zealand's silver lining

A large population of New Zealand found the lockdown to be a positive experience. <http://digit.in/may21-24>

An atoll in the Pacific ocean



When India collided with Eurasia

Magnetism in Himalayan rocks is allowing geologists to figure out the exact sequence of events that took place when India collided with Eurasia.

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If a time traveler keeps skipping into the past at regular intervals with a compass, there are going to be times when the compass needle points to the south pole. Scientists can dig the rocks for samples, to investigate the history of geomagnetic reversals, which is then used as a chronobiological tool to figure out a number of things about

the history of the planet, such as when and where continental plates moved. Embedded within the rock are magnetic minerals that orient themselves in the direction of the north pole at the time of formation. Paleomagnetism is the study of this record in the rocks, of the history of Earth's magnetic field. The magnetism of the minerals in the rocks inform the researchers of the latitude in which the rock was formed, and the Uranium-lead dating of the zircon in the rocks allows them to pinpoint the time of formation. This in turn allows researchers to figure out how continental plates moved and

interacted in the past. Paleomagnetic investigations of the rocks in the Himalayas is allowing scientists to finally figure out the exact sequence of events that took place when the Indian plate collided against the Eurasian plate.

The conventional story is pretty straightforward. The Indian Plate started on the journey northwards, after breaking away from Gondwana, a continent near the south pole that included Africa and South America, around 225 million years ago. There wasn't an Indian Ocean yet, and the Indian plate travelled through the Tethys Ocean. Along the way, around a 100 million years ago, Madagascar broke away from India. Finally, around 50 million years ago, the Indian Plate collided with the Asian plate, sliding under it into the mantle in a process known as subduction, while the Himalayas began to be formed above. The Himalayas are an imposing climatic barrier trapping clouds and directing winds, with a number of major rivers originating in the Tibetan Plateau and the Himalayas. Everything from the climate to ocean chemistry is affected by the Himalayas, and researchers can have a better understanding of the role that the Himalayas has to play in the global climate, by understanding its formation more precisely.

However, there were many aspects of this story that scientists have not agreed on. The earliest date of the collision goes as far back as 90 million years ago, while the latest date is 35 million years ago. The amount of continental crust that got subducted is not agreed on, and neither is the thickness of both the interacting plates. Researchers are getting an increasing understanding of the rate at which the Himalayas are rising, and the angle at which the Indian plate is moving under the Asian plate, and at what rate. Computer models used to simulate the sequence of events from what is known about the collision makes the Indian plate move too fast in too little time. Researchers